CBSE CLASS X • INFORMATION TECHNOLOGY (SUBJECT CODE 402)

UNIT 2: ELECTRONIC SPREADSHEET (ADVANCED) — CHAPTER 7
SHARE AND REVIEW A SPREADSHEET

Comprehensive Detailed Theory • Unabridged Practical Procedures • Conflict Resolution • Track Changes & Merging • Solved Board PYQs

---

1. Concept of Collaboration & User Registration

In large corporate offices, academic institutions, and multinational businesses, teams frequently work on identical spreadsheet files simultaneously. For instance, multiple regional correspondents entering sales responses across various cities, or financial managers collaborating on quarterly budgeting. When multiple users view, modify, and review the same file simultaneously, it is termed **Document Collaboration**.

User Registration (Prerequisite for Collaborative Auditing):

Before collaborating or recording edits, each user must register their profile in LibreOffice. This ensures that Calc can automatically attribute every recorded modification and inserted comment to the specific author with an exact timestamp.

Step 1: Navigate to the Menu bar and click **Tools** → **Options...** (or press **Alt + F12**).

Step 2: In the left panel, expand the **LibreOffice** group and click on **User Data**.

Step 3: In the right pane, fill in user details: *Company*, *First/Last name/Initials*, *Street*, *City/State/Zip*, *Country/Region*, and *Title/Position*.

Step 4: Ensure the checkbox **'Use data for document properties'** is checked.

Step 5: Click **Apply**, then click **OK**. All subsequent edits and comments will carry this user identity.

2. Setting Up, Opening & Saving a Shared Spreadsheet

Definition: Sharing a spreadsheet allows multiple users to access, enter data, and edit the same workbook concurrently over a local network or shared cloud folder.

A. Setting Up a Spreadsheet for Sharing (Unabridged Procedure):

Step 1: Open the spreadsheet workbook you intend to share.

Step 2: Choose **Tools** → **Share Spreadsheet...** from the Menu bar. The *Share Document* dialog box opens.

Step 3: Check the checkbox labelled **'Share this spreadsheet with other users'**. (Clearing this box disables sharing and restores exclusive single-user mode).

Step 4: The bottom pane (*Users Currently Accessing This Spreadsheet*) lists all users currently connected, their access status, and login timestamps.

Step 5: Click **OK**. If the document has unsaved edits, Calc prompts: *'The spreadsheet must be saved to begin sharing. Do you want to continue?'*. Click **Yes**.

Step 6: Title Bar Confirmation: Once shared mode is active, Calc displays the word **(shared)** on the title bar immediately following the document name (e.g., CompanyBudget.ods (shared) - LibreOffice Calc).

B. Opening a Shared Spreadsheet:

When opening a shared document (via File → Open or Ctrl + O), Calc displays an informational notification banner stating: *'The spreadsheet is in shared mode. This allows multiple users to access and edit the spreadsheet at the same time. Changes to formatting attributes... and some functionalities like editing charts and drawing objects are not available in shared mode.'*. Click **OK** to proceed into shared mode.

C. Features Allowed vs. Disabled in Shared Workbooks (CBSE Exam High-Yield):

To prevent data corruption during concurrent editing, Calc disables several structural features in shared mode:

Category	Allowed / Supported Features	Disabled / Prohibited Features in Shared Mode
Data Entry & Edits	Entering text, numbers, formulas, dates; editing cell contents; deleting cell values.	Conditional formatting; Data Validation (Validity); Define Range; Subtotals; Consolidate; Multiple Operations.
Formatting	Basic direct formatting (font size, styles) may be typed, but custom formatting attributes are restricted.	**Merging cells (Format → Merge Cells)**; formatting print ranges.
Objects & Insert	Viewing existing charts and drawings.	**Inserting pictures/graphs**; inserting charts; OLE objects; floating frames; audio/video media; Fontwork; signature lines.
Sheet Architecture	Switching between existing sheets; viewing tabs.	Insert Sheet from File; Rename Sheet; Delete Sheet; Move/Copy Sheet; change Sheet Tab Color.

A.S. NOTES IT-402

Review & Tracking	**Merge Document** (merging changes from another file).	Edit → Track Changes (Record, Show, Manage are disabled inside a shared file); inserting comments.

D. Saving a Shared Spreadsheet & Conflict Resolution:

When saving a shared workbook (File → Save or Ctrl + S), Calc evaluates concurrent edits through two scenarios:

- **Case 1: No Conflicts (Non-overlapping Changes):** If other users modified cells in different areas (e.g., User 1 modified *A10:F30* in Sheet1 while User 2 modified *A55:E80*), Calc updates the file, saves your work, and displays a notification: *'Your spreadsheet has been updated with changes saved by other users.'*. Modified cells saved by others are visually highlighted with a **red border**.
- **Case 2: Conflicting Changes (Overlapping Edits):** If multiple users modified the *exact same cell(s)* with differing data, the **Resolve Conflicts** dialog box opens automatically.
- **Conflict Resolution Options:** The saving user must review each conflict and choose:
  - **Keep Mine:** Retains your newly entered cell value and overwrites the other user's change.
  - **Keep Other:** Discards your edit and preserves the value saved by the other user.
  - **Keep All Mine / Keep All Others:** Resolves all remaining cell collisions in one action.
  - *File Locking Rule:* While conflict resolution is in progress, no other user can save the shared file (they receive a notification that the file is temporarily locked due to a merge-in).

3. Recording Changes in a Spreadsheet (Audit & Review Mode)

Definition: Recording Changes is a specialized document auditing mode that logs every insertion, deletion, and content edit as a tentative modification, allowing authorized managers or teachers to inspect, verify, accept, or reject each change before making it permanent.

Prerequisites Before Enabling Record Changes:

- **Sharing Must Be Disabled:** The Record Changes feature is disabled while a workbook is in active shared mode. Users must first turn off sharing via Tools → Share Spreadsheet... (uncheck the box) and save the document.
- **Confirm Sharing is Disabled:** Confirm that the word **(shared)** is no longer present in the title bar after the filename.

Unabridged Step-by-Step Procedure to Record Changes:

Step 1: Open the non-shared spreadsheet document.

Step 2: Choose **Edit** → **Track Changes** → **Record** from the Menu bar (or press **Ctrl + Shift + C**). A checkmark appears beside *Record* indicating tracking is active.

Step 3: Perform your edits, data updates, or corrections in the worksheet cells.

Step 4: Visual Cell Indicator (CBSE Must-Know): Around every edited cell, Calc automatically draws a **coloured border with a dot in the upper left-hand corner**.

Step 5: Hover Tooltip: Hovering the mouse pointer over the edited cell displays an automatic yellow tooltip describing the author, modification date, time, and previous value (e.g., *'pace2, 29/07/2026 15:30:44: Cell C7 changed from 136 to 236'*).

Step 6: (Optional) To protect the review process from being turned off by unauthorized users, choose **Edit** → **Track Changes** → **Protect...** and enter a security password.

4. Adding, Formatting & Editing Comments in Changed Cells

While Calc generates an automatic comment recording raw value changes, reviewers frequently need to append qualitative explanations (e.g., explaining why budget figures were increased).

A. Adding an Explanatory Change-Comment:

Step 1: Select the edited cell (the cell displaying the coloured border and corner dot).

Step 2: Choose **Edit** → **Track Changes** → **Comment...** from the Menu bar (or press **Ctrl + Alt + C**).

Step 3: The *Comment* dialog box appears. The title bar displays the permanent system-generated description (which cannot be changed).

Step 4: In the **Text** area, type your explanatory note (e.g., *'Class strength increased by 15%; revised budget submitted for approval.'*).

Step 5: (Optional) Use the left and right navigation arrows to step through other changed cells and add comments.

Step 6: Click **OK**.

B. Formatting and Editing Comments:

To customize the appearance of a comment box: right-click the cell containing the comment and select **Edit Comment**. Highlight the text inside the comment box and use the **Formatting Toolbar** to modify font style, font size, text color, highlight fill, or alignment. To edit the text, use Edit → Track Changes → Comment..., modify the message, and click OK.

D.A.V. NOTES IT-402

5. Reviewing, Filtering & Accepting/Rejecting Changes

Once all reviewers complete their entries, the project head, manager, or teacher reviews the modifications to finalize the official document.

A. Filtering Tracked Changes (Show Changes Dialog):

Choose **Edit** → **Track Changes** → **Show...**. The *Show Changes* dialog allows filtering visible edits using specific parameters:

- **Date:** Filters edits by timeline (e.g., earlier than, since, between specific dates/times).
- **Author:** Isolates modifications made by a specific reviewer (essential when auditing multi-user contributions).
- **Range:** Restricts displayed markup to a specific rectangular cell block (e.g., *Sheet1.B4:E20*).
- **Comment:** Searches text strings within attached comments.
- **Show accepted / Show rejected changes:** Checkboxes to display historical decisions.

B. Managing Changes (Accepting or Rejecting Edits):

Step 1: Open the reviewed document and choose **Edit** → **Track Changes** → **Manage...**. The *Manage Changes* dialog box appears.

Step 2: The dialog displays a tabular list of all uncommitted edits showing **Action**, **Position** (sheet/cell), **Author**, **Date/Time**, and **Comment**.

Step 3: Click on any entry in the list; Calc immediately scrolls to and highlights the target cell in the worksheet.

Step 4: Select the desired command button at the bottom of the dialog:
- **Accept:** Commits the selected modification permanently into the spreadsheet.
- **Reject:** Reverts the selected cell back to its original value prior to tracking.
- **Accept All:** Approves every listed change in a single action.
- **Reject All:** Discards all modifications and restores the entire original dataset.

Step 5: Click **Close** when finished, and save the finalized document.

6. Merging and Comparing Spreadsheets

In distributed workplace workflows, multiple reviewers often receive identical copies of a file and return separate edited versions independently (e.g., two managers, Mr. Raj and Ms. Meera, editing separate budget copies: Budget_Raj.ods and Budget_Meera.ods). Calc provides two dedicated features to consolidate them:

A. Merging Worksheets (Consolidating Multiple Reviews):

Why Merge? Merging combines edits from multiple reviewed copies into the original document in one go, enabling the owner to review all contributors' suggestions simultaneously rather than opening files one by one.
Prerequisite: All edited copies *must have had Record Changes enabled* during editing.

Step 1: Open the master original spreadsheet (e.g., *Budget_Original.ods*).

Step 2: Navigate to the Menu bar and choose **Edit** → **Track Changes** → **Merge Document...**.

Step 3: A file selection dialog opens. Browse to locate the first reviewed file (e.g., *Budget_Raj.ods*) and click **Open**.

Step 4: Calc merges the recorded edits into the master file. The *Manage Changes* dialog appears displaying all incoming edits.

Step 5: Multiple Reviewer Indicator (+): If multiple reviewers modified the exact same cell, Calc displays a **plus (+) sign** next to that entry in the list. Clicking the **+** sign expands the row to display each reviewer's proposed value.

Step 6: Accept or reject the changes as required, repeat for any other reviewer files (e.g., *Budget_Meera.ods*), and save the merged document.

B. Comparing Documents (When Reviewers Forgot to Track Changes):

Why Compare? If reviewers made changes to a workbook copy *without turning on Record Changes*, Calc can compare the modified copy against the original document to identify and list all differences automatically.

Step 1: Open the modified/edited spreadsheet file (e.g., *Rev_Bill.ods*).

Step 2: Choose **Edit** → **Track Changes** → **Compare Document...** from the Menu bar.

Step 3: In the file selection dialog, browse to select the original base document (e.g., *Bill.ods*) and click **Open**.

Step 4: Calc compares the two files cell-by-cell and automatically generates the *Manage Changes* dialog box. Calc categorizes differences as follows:
- **Inserted:** Data present in the edited file but absent in the original.
- **Deleted:** Data present in the original file but missing from the edited file.
- **Changed:** Cells where contents or numbers differ between the two files.

Step 5: Review each identified discrepancy, click **Accept** or **Reject**, and save the updated file.

AS NOTES IT-402

7. CBSE Board Examination Question Bank & Model Marking Scheme Solutions

Part A: Objective & Multiple Choice Questions (1 Mark Each)

Q1. Reviewers and authors can add their _____ to explain their changes in the cell of Spreadsheet. [CBSE SQP 2022-23]
(a) Comments (b) Hyperlink (c) Worksheet (d) Macros

Ans: (a) Comments (accessible via Edit → Track Changes → Comment... or Ctrl + Alt + C).

Q2. Sahil and his partners have created a list of their company employees in one of the spreadsheets in the office. Now the rest of the task they want to complete from home. Which option should they use to access the same Spreadsheet from home? [CBSE SQP 2023-24 / Board Paper 2]
(a) Consolidate Worksheet (b) Link Worksheet (c) Merge Worksheet (d) Lock Worksheet

Ans: (b) Link Worksheet (or *Shared Spreadsheet* feature for concurrent multi-user editing).

Q3. Spreadsheet software can find the changes by _____ sheets. [CBSE SQP 2023-24]
(a) Changing (b) Comparing (c) Renaming (d) Editing

Ans: (b) Comparing (using Edit → Track Changes → Compare Document...).

Q4. A colored border, with _____ appears around a cell where changes are made in a shared worksheet. [CBSE Board 2023]
(a) a dot in the upper left-hand corner (b) a dot in the lower left-hand corner
(c) a cross in the upper left-hand corner (d) a cross in the upper right-hand corner

Ans: (a) a dot in the upper left-hand corner

Q5. _____ feature of Calc helps to track what data was changed, when the change was made, who made the change and in which cell the change has occurred. [CBSE Board 2023 / Board 2024]
(a) Record Changes (b) Edit record (c) Change record (d) Macro

Ans: (a) Record Changes (Path: Edit → Track Changes → Record).

Q6. Which Calc feature allows multiple users to suggest edits while preserving the original data? [CBSE Board Paper 1 (Set 4)]
(A) Protect Sheet (B) Share Document (C) Track Changes (D) Insert Comments

Ans: (C) Track Changes (allows suggesting temporary edits that can be accepted or rejected by the author).

Q7. _____ option of Calc helps in viewing the changes made to a worksheet using original and edited worksheet. [CBSE Board 2024]
(A) Edit → Revise document (B) Edit → Consolidate
(C) Edit → Compare document (D) Edit → Collaborate document

Ans: (C) Edit → Compare document

Q8. In Calc, shared workbooks allow: [CBSE Board 2024]
(A) Merging cells (B) Conditional formatting (C) Inserting pictures/graphs (D) Adding text

Ans: (D) Adding text (Merging cells, conditional formatting, and inserting pictures/graphs are strictly disabled in shared mode).

Q9. Which menu path in LibreOffice Calc is used to accept or reject changes made in a shared spreadsheet? [CBSE SQP 2025-26 / 2026-27]
(a) Tools > Review Changes > Manage (b) File > Options > Track Changes
(c) Edit > Track Changes > Manage (d) View > Track Changes > Review

Ans: (c) Edit > Track Changes > Manage

Q10. A company's finance team is preparing the quarterly budget report. Two managers, Mr. Raj and Ms. Meera, work on the same spreadsheet file separately (Budget_Raj.ods and Budget_Meera.ods). Now, before submitting the final report to the CEO, the team needs to: 1. Check differences in both versions; 2. Merge approved changes into one final file. As the data entry operator, which steps will you follow in LibreOffice Calc? [CBSE SQP 2024-25 / 2025-26 / 2026-27]
(a) Directly open Budget_Raj.ods and overwrite it with Budget_Meera.ods.
(b) Open Budget_Raj.ods, use Compare Document to review differences, then use Merge Document to finalize.
(c) Open Budget_Meera.ods and delete conflicting data manually.
(d) Use only Merge Document, since it automatically compares files too.

Ans: (b) Open Budget_Raj.ods, use Compare Document to review differences, then use Merge Document to finalize.

Part B: Short Answer Questions (2 Marks Each)

Q11. What are the benefits of sharing a spreadsheet? [CBSE SQP 2024-25, Q.13]

Ans:
1. **Concurrent Collaboration:** Enables multiple team members across different locations to input, edit, and audit data simultaneously in the same workbook.
2. **Efficiency & Data Integrity:** Eliminates redundant versions, manual copy-pasting between separate files, and delays in compiling organizational records.

DAV NOTES IT-402

Q12. (a) What must you do before enabling the Record Changes feature?
(b) How can you confirm that sharing has been disabled in the spreadsheet? [CBSE SQP 2025-26 / 2026-27, Q.12]

Ans:
(a) You must **disable the sharing feature** of the spreadsheet (via Tools → Share Spreadsheet..., uncheck *'Share this spreadsheet with other users'*, and click OK/Save) because Record Changes cannot be activated in shared mode.
(b) You can confirm that sharing is disabled by checking the **Title Bar** at the very top of the window; the word **'(shared)'** will no longer appear next to the document filename.

Q13. Why do we need to merge worksheets? [CBSE Board 2023, Q.13]

Ans: We need to merge worksheets when multiple reviewers have independently made edits or recorded changes on separate copies of the same workbook. Merging brings all recorded modifications into a single master sheet at once, allowing the author to review, compare alternative proposals, and accept or reject changes in one consolidated view.

Part C: Practical Procedural & Case-Study Questions (4 Marks Each)

Q14. Mr. Prasanna, a General Manager in a multinational company, has been given two spreadsheets — an original billing file Bill.ods and a reviewed version of the same spreadsheet Rev_Bill.ods. Both contain customer billing details prepared by different teams. Write the steps he should follow to compare these worksheets and identify the differences between them. [CBSE Board Paper 1 (Set 4), Q.18]

Ans: To compare the worksheets and identify all differences, Mr. Prasanna should execute the following steps in LibreOffice Calc:
Step 1: Open the reviewed billing spreadsheet file (Rev_Bill.ods) in LibreOffice Calc.
Step 2: Navigate to the Menu bar and choose **Edit** → **Track Changes** → **Compare Document...**.
Step 3: An *Open Document* dialog box will appear. Browse to locate and select the original billing file (Bill.ods), then click **Open**.
Step 4: Calc compares the two files cell-by-cell and opens the **Manage Changes** dialog box.
Step 5: In the Manage Changes dialog, all differences are categorized and listed (as *Inserted*, *Deleted*, or *Changed*) with author, cell coordinates, date, and description.
Step 6: Mr. Prasanna can select each discrepancy individually to inspect it on the sheet and click **Accept** or **Reject** (or click *Accept All* / *Reject All*) to finalize the billing records, then save the workbook.

A.S. NOTES IT 402